



Recognition of Pes Cavus Foot Using Smart Insole: A Pilot Study

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Abstract. The presence of pes cavus, a high-arched foot, is a potential reason for some neuromuscular problems. Active research efforts are being made to devise portable systems for monitoring and early detection of foot deviations. In line with that, we have developed instrumented insoles that incorporate force and inertial sensors and used them to capture data from sixty-four subjects; among them, there were forty-four subjects with normal feet arches and twenty subjects exhibiting pes cavus. We applied a 1D convolutional neural network to extract features and classify data. The trained model allowed for a recognition rate of more than 96%. The presented use case could inspire further research on using smart footwear for pes cavus screening and progression monitoring.

Keywords: Pes cavus · Smart insole · Plantar pressure · 1D CNN

1 Introduction

Individuals with pes cavus exhibit a higher foot arch compared to those with normal feet. Surveys show that pes cavus is present in around 8–15% of the world population [1]. This condition leads to neuromuscular problems, including unstable gait, foot pain, and ankle sprains [1, 2]. Its early recognition and appropriate assessment are essential for successful treatment and management [2]. So far, morphological parameters were always used to serve the classification of the foot type. Those were obtained using techniques of visual non-quantitative inspection, anthropometric measurement, foot-print parameter measurement, and radiographic evaluation [3]. However, these methods are to be practiced by podiatrists and require repetitive training.

Efforts are being made for devising instruments and methods to automate data capture and foot function analysis. The evaluation of the foot locomotion characteristics

is then performed based on kinematic and kinetic parameters. Existing instrumentation includes pressure distribution measurement systems and 3D force plates for obtaining kinetic parameters, and inertial sensors or visual capturing systems to retrieve kinematic ones. However, these systems require time-consuming procedures, and many of them are not applicable to serve regular evaluation or are not accessible to the general public due to high costs. The desire is to introduce portable measurement devices that could support foot type evaluation. As pes cavus progresses with the time, such automated devices could allow for regular screening and early detection of foot abnormalities [21]. Each foot type is associated with a specific pattern of plantar pressure distribution during the gait cycle [5]. In particular, the peak of vertical ground reaction force and the footprint areas during the stance phase were found different between a flat foot and pes valgus [4]. Also, the peak pressures under the five metatarsophalangeal joints and the minimum velocity of the center of pressure during the terminal stance phase [6] differ between a normal foot and pes cavus [5, 6]. The maximum force under the medial arch, the force-time integral under the hallux, the pressure-time integral under the fifth metatarsal head and their normalized values are also different between a normal foot and pes cavus [10]. As to automated methods for foot type classification, so far, artificial neural networks, adaptive neuro-fuzzy inference systems, and K-means algorithms were applied [14–16].

As inertial sensors are easy to incorporate into wearable devices, they can serve gait analysis and applications involving gait and activity recognition [11, 12, 22]. Locomotion characteristics of pes cavus also were explored using accelerometers. Ledoux et al. [7] studied the difference in the calcaneal acceleration at heel strike between a normal foot and pes planus using an accelerometer. Pes cavus foot exhibited a higher acceleration in comparison with the normal foot [9]. Ogon et al. [8] found that acceleration amplitudes and angular rates of subjects with pes cavus are lower compared to respective ones of subjects with pes planus. Based on such differences, recognition systems could be constructed.

However, to date, no classifier was demonstrated that could effectively recognize pes cavus based on the combination of simultaneously captured inertial data and/or single point force sensors. To address this gap, we used data from a custom prototype incorporating inertial and force sensing. We hypothesized that the combination of simultaneously recorded multimodal sensor data could reflect better the specifics of pes cavus and allow for better recognition. Deep learning has been successfully used in different subdomains of biomedical data processing, including image processing, speech recognition, and natural language processing. Thus, we applied a 1D convolutional neural network (1D CNN) to extract gait features for the recognition of pes cavus from data acquired using our prototype.

2 Data Collection

2.1 Smart Insole Implementation

The sensor insole prototype consists of a Bluetooth Low Energy-enabled electronic control module, which connects to nine force sensors, and has an integrated inertial

sensor. The module is based on a wireless system-on-chip of type nRF52832, and contains an inertial sensor of type BMI160 incorporating triaxial accelerometer and gyroscope. Selected force sensors are of type FlexiForce A301 and were attached to a specially designed thin and flexible printed circuit board, with locations matching the main weight-bearing positions of the foot, namely, the hallux, the five metatarsus heads, the lateral foot arch, the medial and lateral heel areas. All data were sampled at 100 Hz and transferred wirelessly to a personal computer. Each control module was attached on the top of the frontal part of the shoe. An assembled instrumented shoe is shown in Fig. 1a.

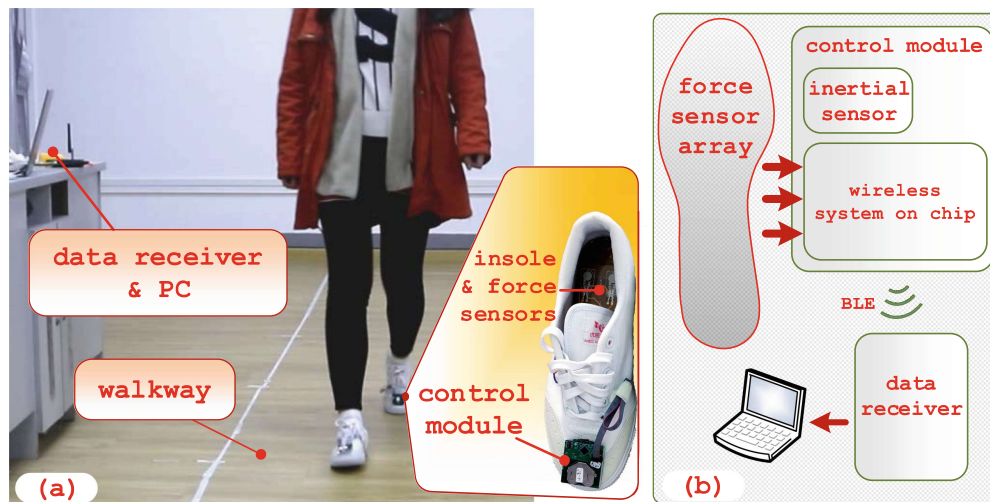


Fig. 1. (a) our smart insole prototype (b) experimental setting.

2.2 Experiment Configuration

Sixty-four volunteers participated in the experiment. Of them, forty-four with normal feet and twenty with pes cavus (Table 1). A podiatrist determined the foot type of each subject. Before executing the experiment, each volunteer was informed about the experimental procedure, signed an informed consent form, and filled a questionnaire about demographic characteristics, general health, and feet status. The data collection procedure was conducted in accordance with the Declaration of Helsinki and was approved by the ethics committee of the University of Technology of Chengdu. To perform a trial, the subject had to walk at a self-selected speed straight ahead for seven meters, wearing instrumented shoes of an appropriate size. Sudden changes in gait were not allowed. Each subject performed twenty trials. Each trial was recorded by a camera. Trial recordings which had missing data due to wireless transmission losses were excluded from the further analysis. Thus, 1130 records of right foot were processed. The characteristics of the subjects are presented in Table 1. The experimental configuration is shown in Fig. 1(b). A representative example of the waveforms of signals captured during a trial is provided in Fig. 2.

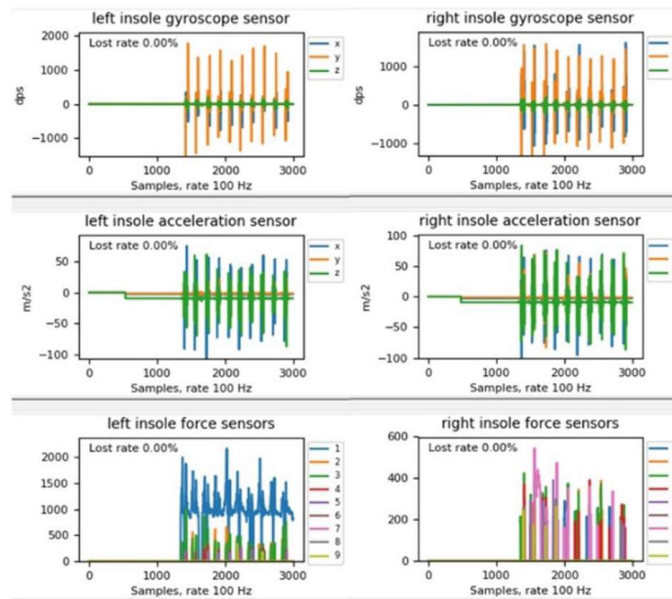


Fig. 2. Waveforms of signals captured during a trial.

Table 1. Characteristics of subjects

Foot type	Gender (male/female)	Height (cm $\pm$ SD)	Weight (kg $\pm$ SD)	Foot size (No ¹ $\pm$ SD)
Normal (44)	31/13	169.1 $\pm$ 8.3	61.7 $\pm$ 10.3	40.1 $\pm$ 2.6
Pes cavus (20)	15/5	169.9 $\pm$ 5.9	61.0 $\pm$ 9.2	39.8 $\pm$ 2.2
Total (64)	46/18	169.3 $\pm$ 7.6	61.6 $\pm$ 10.7	40.0 $\pm$ 2.5

¹Shoe size according to the Continental European system

3 1D CNN

Classical methods for foot type classification are always based on hand-crafted features extracted from plantar pressure. Possibly, these features do not fully reflect the characteristics of the different foot types. Consequently, the classifier may not reach the best performance because the most discriminative features were not found. Deep learning-based methods have a strong ability to learn features. For instance, 1D CNN has been successfully applied to human action recognition [11, 12]. Also, anterior cruciate ligament deficiency was recognized based on plantar pressure data passed to a 2D CNN [13].

In the present work, a 1D CNN deep learning method was used for feature learning and classification of the two types of foot. The typical CNN has a structure that involves convolution layers, a pool layer, and fully connected layers. The stack of convolution and pooling layers allows the automatic feature extraction, from low-level to high-level representations.

Because of the gait variability, the trials of the same subject differed in length; trials of different subjects also differed in length. To obtain time series with equal lengths, shorter series were zero padded to the length of the longest one. The structure of the 1D CNN used for the foot type classification in the present study is shown in Fig. 3. The network includes four convolutional layers, four batch-normalized layers, four activation layers, three max pool layers, one flatten layer, one dropout layer, and three fully connected layers. Each convolution layer and the first max pool layer have a filter size of 1×7 , and the second max pool layer has a filter size of 1×7 . Compared with filters of larger sizes, smaller-size ones have a lower number of parameters to learn, and they can also learn features with higher complexity. Thus, two convolution layers are stacked before the max-pooling layers [17]. Batch normalization and dropout were used to shorten the training time and improve the performance of the network [18, 19]. We used ADADELTA optimizer. It is an extension of ADAGRAD, and with the progression of the learning, the learning rate decreases monotonically [20].

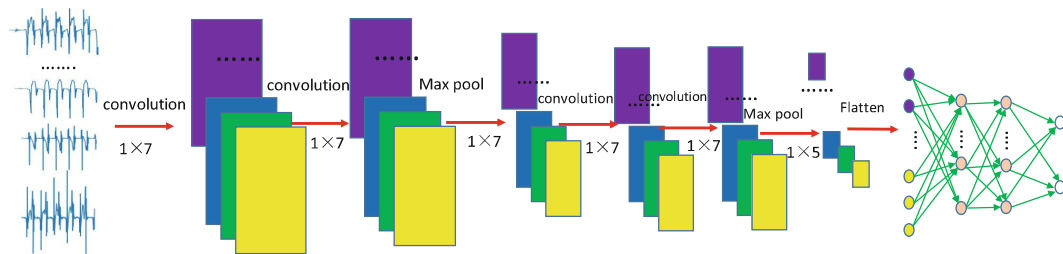


Fig. 3. The architecture of the proposed 1D CNN. It consists of two convolution layers, one max pool layer, two convolutional layers, a max pool layer, one flatten layer, and three fully connected layers. After every convolutional layer, there is a batch normalization layer.

4 Results

We tested the described network with the inertial and force sensor data separately, as well as taken together. To train the network, 80% of the samples were used, while the rest of them served for the testing. We trained the 1D CNN with 100 training epochs. The accuracies obtained by the 1D CNN are listed in Table 2. The best accuracy was obtained in two cases - when all modalities were used, and also for force data only. The training and validation losses are shown in Fig. 4. The confusion matrices are shown in Fig. 5.

For comparison with traditional methods, a random forest classifier with default parameter settings was tested as it is the best one to classify gait-related features [23]. The classifier took calculated input features. These were the maximum, minimum, mean, range, zero crossing rate, root-mean-square value, variance, standard variance, skewness, and kurtosis, obtained for each series. In Table 3, the performance of the

random forest classifier is given. When applied to both inertial and force data, the classifier has shown the best accuracy. When using single-modality data, the accuracy of the classifier using force data was better than one only using inertial data.

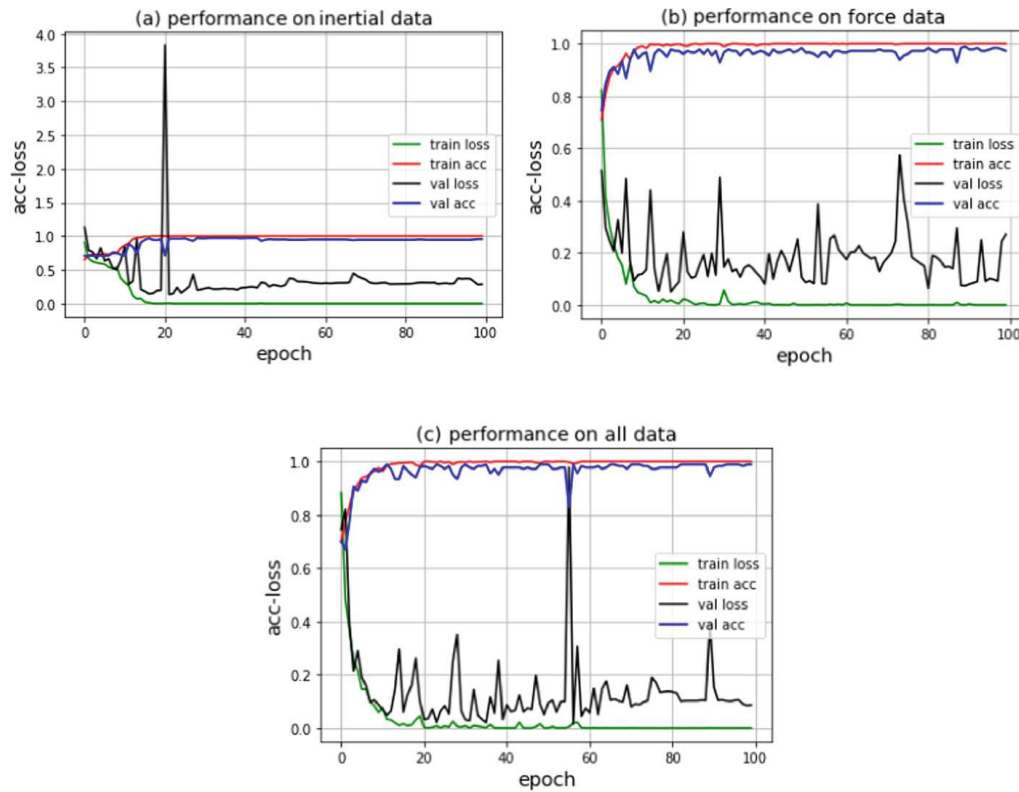


Fig. 4. Accuracy and loss for three kinds of sensor data passed during each epoch. (a) accuracy and loss on inertial data, (b) accuracy and loss on the force data, (c) accuracy and loss on all data.

Table 2. Performance of the 1D-CNN

	Foot type	Inertial data	Force data	All data
Accuracy	Normal	99.44%	100%	100%
	Pes cavus	85.42%	100%	100%
	Total	96.46%	100%	100%

Table 3. Performance of the random forest classifier

	Foot type	Inertial data	Force data	All data
Accuracy	Normal	98.31%	97.75%	99.42%
	Pes cavus	75%	83.33%	86.79%
	Total	93.36%	94.69%	96.46%

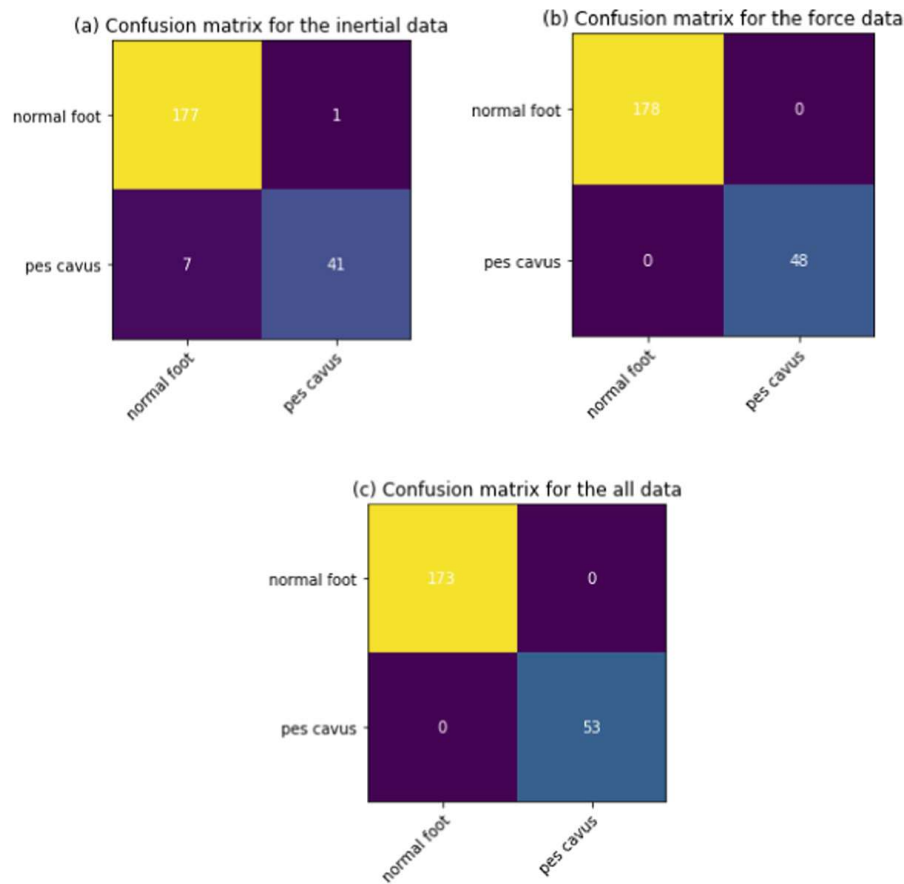


Fig. 5. Confusion matrix of foot type classification for (a) inertial data, (b) force data and (c) all data.

5 Discussion and Conclusion

The 1D CNN demonstrated high performance in the processing of smart insole data. The accuracies for the test data set of force data and all data reached 100%, respectively. Among the three tested variants, the force data and full data showed the best performance; this result could be evidence that the plantar pressure distribution pattern is a reliable indicator of the foot arch abnormalities. Inertial data also proved to be suitable for pes cavus detection. To date, there is no parameter described that can serve for the evaluation of pes cavus and flat foot at heel strike [7, 9]. Our results from inertial data classification indicate that the 1D CNN was able to find discriminative parameters. Even though only a single inertial sensor was used, the accuracy in discrimination between the two types of foot reached 96.46%. Increasing the number of sensors and improving sensor placement may allow for better accuracy. When we pooled inertial data and force data, the achieved accuracy was equal with the one from force data only. Multi-modal features could serve for the reliable discrimination of the foot types. As to the random forest method with handcrafted features, its performance was not as good as the one of 1D CNN for all the three data inputs.

Overall, the 1D CNN proved to be an appropriate choice for processing force and inertial sensor data. In the application of foot type classification, it allowed for reaching a good discriminative performance. Results imply that the smart insole, in combination with the proposed method, can be used for screening and progression monitoring of pes cavus.

Acknowledgments. This project was supported in parts by the Key Project 2017GZ0304 of the Science and Technology Department of Sichuan province and the Key Program of Joint Funds of the National Natural Science Foundation of China, grant U1505251.

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